MATH 342W / 642 / RM742 Spring 2026 HW #1

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Problem 1

These are questions about Silver's book, the introduction and chapter 1.

(a) [easy] What is the difference between *predict* and *forecast*? Are these two terms used interchangeably today?

According to Silver, a "prediction" as used by Shakespeare was akin to prophecy or fortune telling, interpreting signs "as to gain an advantage from them".

A "forecast on the other hand was obtained by utilizing available information and human judgment, planning under uncertainty, to obtain an idea of what might occur.

(b) [easy] What is John P. Ioannidis's findings and what are its implications?

Ioannidis published a paper in which he detailed the fact that most published scientific research findings are false. He wrote about how predictions that were proven successful in lab testing were likely to fail when applied to real life. This hypothesis was later proven true by attempts to replicate the positive findings.

The implication is that many reported successes and seemingly confirmed predictions can be the result of small sample sizes, environments tailored to support theories, and other undermining factors that expresses a need for skepticism.

(c) [easy] What are the human being's most powerful defense (according to Silver)? Answer using the language from class.

Human beings' most powerful defense according to silver is the ability to detect patterns: probabilistic thinking.

(d) [easy] Information is increasing at a rapid pace, but what is not increasing?

Human understanding of how to properly use and translate that information into applicable knowledge.

(e) [difficult] Silver admits that we will always be subjectively biased when making predictions. However, he believes there is an objective truth. In class, how did we describe the objective truth? Answer using notation from class i.e. $t, f, g, h^*, \delta, \epsilon, t, z_1, \ldots, z_t, \delta, \mathbb{D}$, $\mathcal{H}, \mathcal{A}, \mathcal{X}, \mathcal{Y}, X, y, n, p,\ x_{\cdot 1}, \ldots, x_{\cdot p}, x_{1\cdot}, \ldots, x_{n\cdot}$, etc.

(f) [easy] In a nutshell, what is Karl Popper's (a famous philosopher of science) definition of *science*?

According to Karl Popper, the ability to make testable predictions that can be proven wrong by real world testing.

(g) [harder] Why did the ratings agencies say the probability of a CDO defaulting was 0.12% instead of the 28% that actually occured? Answer using concepts from class.

(h) [easy] What is the difference between *risk* and *uncertainty* according to Silver's definitions?

Risk is the situation in which outcomes have probabilities or odds that are known and can be accounted for.

Uncertainty is risk that is difficult to measure, a situation where those probabilities and odds cannot be easily estimated, where the ability to predict outcomes is difficult.

(i) [difficult] How does Silver define *out of sample*? Answer using notation from class i.e. $t, f, g, h^*, \delta, \epsilon, z_1, \ldots, z_t, \delta, \mathbb{D}, \mathcal{H}, \mathcal{A}, \mathcal{X}, \mathcal{Y}, X, y, n, p, x_{\cdot 1}, \ldots, x_{\cdot p}, x_{1\cdot}, \ldots, x_{n\cdot}$, etc. WARNING: Silver defines *out of sample* completely differently than the literature, than practitioners in industry and how we will define it in class in a month or so. We will explore what he is talking about in class in the future and we will term this concept differently, using the more widely accepted terminology. So please forget the phrase *out of sample* for now as we will introduce it later in class as something else. There will be other such terms in his book and I will provide this disclaimer at these appropriate times.

Silver uses out of sample to mean that the data-generating process $y = t(z_1, \ldots, z_t)$ has changed, so the historical dataset D is no longer representative.

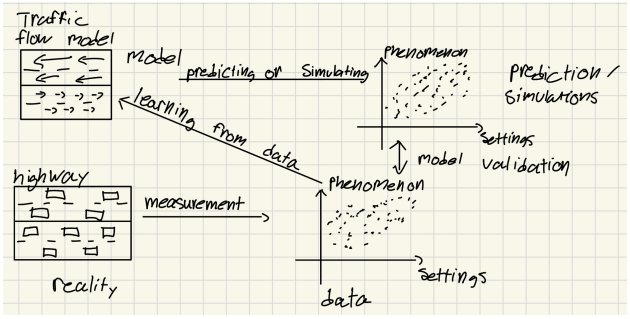
(j) [harder] Look up *bias* and *variance* online or in a statistics textbook. Connect these concepts to Silver's terms *accuracy* and *precision*. This is another example of Silver using non-standard terminology.

Bias occurs when the expected value of an estimator does not equal the true value, while variance measures how spread out estimates are around their mean. Silver's term *accuracy* corresponds to low bias, since it describes how close predictions are to the true value. His term *precision* corresponds to low variance, since it describes how consistent the predictions are.

Problem 2

Below are some questions about the theory of modeling.

(a) [easy] Redraw the illustration of Earth and the table-top globe except do not use the Earth and a table-top globe as examples (use another example). The quadrants are connected with arrows. Label these arrows appropriately.



(b) [easy] Pursuant to the fix in the previous question, how do we define *data* for the purposes of this class?

Data is measurements that is observed from reality, based on the phenomenon that is of interest.

(c) [easy] Pursuant to the fix in the previous question, how do we define *predictions* for the purposes of this class?

Predictions are generated by building a model that learns from the observed data then outputs predictions/simulations.

(d) [easy] Why are "all models wrong"? We are quoting the famous statisticians George Box and Norman Draper here.

Because the metrics, variables, and data used to generate our predictions can be incomplete, and it is nearly impossible to fully encapsulate every factor influencing a phenomenon within any given model.

(e) [harder] Why are "[some models] useful"? We are quoting the famous statisticians George Box and Norman Draper here.

Some models are useful because, while they cannot capture every factor involved, they can capture enough to generate an accurate enough prediction to make decisions off of.

(f) [harder] What is the difference between a "good model" and a "bad model"?

A "good" model is one that is useful for measuring the phenomenon you identify and creating a prediction or simulation based on those measurements, while minimizing error.

A "bad" model is the opposite. It fails to measure the selected phenomenon in a way that can be applied in a useful manner, producing inaccurate predictions.

Problem 3

We are now going to investigate the famous English aphorism "an apple a day keeps the doctor away" as a model. We will use this as springboard to ask more questions about the framework of modeling we introduced in this class.

(a) [easy] Is this a mathematical model? Yes / no and why.

No, there are no numerical inputs or outputs, so it isn't a mathematical model.

(b) [easy] What is(are) the input(s) in this model?

Whether or not a person ate an apple each day.

(c) [easy] What is(are) the output(s) in this model?

The health of the person, leading to whether they would need to see a doctor or not.

(d) [harder] How good / bad do you think this model is and why?

It is not a good model because health depends on far more inputs than simply whether or not an apple is eaten per day.

(e) [easy] Devise a metric for gauging the main input. Call this x_1 going forward.

$x_1 =$ If an apple was consumed on a given day or not

(f) [easy] Devise a metric for gauging the main output. Call this y going forward.

$y =$ How healthy a person is based on doctor visits.

(g) [easy] What is $\mathcal{Y}$ mathematically?

$\mathcal{Y}$ is the output space of the model, where the options are visiting the doctor or not: binary.

(h) [easy] Briefly describe $z_1, \ldots, z_t$ in English where $y = t(z_1, \ldots, z_t)$ in this *phenomenon* (not *model*).

$z_1, \ldots, z_t$ are the causal drivers of human health for this model, things like diet, exercise, etc.

(i) [easy] From this point on, you only observe x_1. What is the value of p?

$p = 1$, because there is only one proxy.

(j) [harder] What is $\mathcal{X}$ mathematically? If your information contained in x_1 is non-numeric, you must coerce it to be numeric at this point.

$\mathcal{X}$ is the input space of the model, also binary, where the options are apples being consumed or not. This can be represented as $X_1 = 0$: no apples consumed, or $X_1 = 1$: apple is consumed.

(k) [easy] How did we term the functional relationship between y and x_1? Is it approximate or equals?

y can only be approximate to the function $f(x_1)$ when there is no error term.

(l) [easy] Briefly describe *superivised learning*.

Supervised learning is an empirical method that uses historical data to learn a mapping from inputs to outputs.

(m) [easy] Why is *superivised learning* an *empirical solution* and not an *analytic solution*?

Because supervised learning is based on historical data to approximate the function instead of deriving it analytically with mathematical terms and formulas.

(n) [harder] From this point on, assume we are involved in supervised learning to achieve the goal you stated in the previous question. Briefly describe what $\mathbb{D}$ would look like here.

$\mathbb{D}$ is the set of historical data (input-output pairs) used to train the model.

(o) [harder] Briefly describe the role of $\mathcal{H}$ and $\mathcal{A}$ here.

$\mathcal{H}$ is the set of candidate functions to approximate the relationship between x and y.

$\mathcal{A}$ is the algorithm that selects the optimal function in $\mathcal{H}$.

(p) [easy] If $g = \mathcal{A}(\mathbb{D}, \mathcal{H})$, what should the domain and range of g be?

The domain is $\mathcal{X}$ and the range is $\mathcal{Y}$

(q) [easy] Is $g \in \mathcal{H}$? Why or why not?

Yes because $\mathcal{A}$ selects g from $\mathcal{H}$

(r) [easy] Given a never-before-seen value of x_1 which we denote x^*, what formula would we use to predict the corresponding value of the output? Denote this prediction $\hat{y}^*$.

$\hat{y}^* = g(x^*)$.

(s) [harder] f is the function that is the best possible fit of the phenomenon given the covariates. We will unfortunately not be able to define "best" until later in the course. But you can think of it as a device that extracts all possible information from the covariates and whatever is left over δ is due exclusively to information you do not have. Is it reasonable to assume $f \in \mathcal{H}$? Why or why not?

No, the function function f is generally too complex to lie in the restricted hypothesis space $\mathcal{H}$. We are trying to get as close as possible to H using f

(t) [easy] In the general modeling setup, if $f \notin \mathcal{H}$, what are the three sources of error? Copy the equation from the class notes. Denote the names of each error and provide a sentence explanation of each. Denote also e and $\mathcal{E}$ using underbraces / overbraces.

$$y = g(x_1, ..., x_p) + \underbrace{\underbrace{(h^* - g)}_{\text{estimation error}} + \underbrace{(f - h^*)}_{\epsilon} + \underbrace{(t - f)}_{\delta}}_{e}$$

Ignorance error δ: error due to missing variables or lacking data.

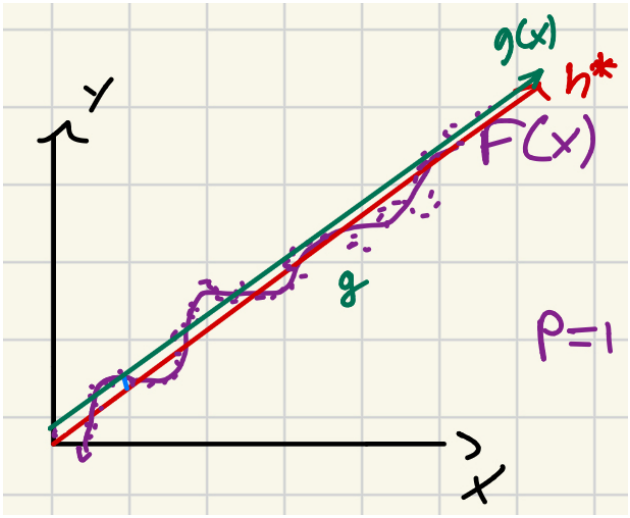
Misspecification error $(f - h^*)$: error caused by restricting the hypothesis space H so that it does not contain the best possible function.

Estimation error $(h^* - g)$: error due to finite data, preventing the algorithm from exactly recovering h^*.

(u) [easy] In the general modeling setup, for each of the three source of error, explain what you would do to reduce the source of error as best as you can.

Ignorance error can be reduced by collecting better proxies, misspecification error by expanding H, and estimation error by collecting more data or improving A.

(v) [harder] In the general modeling setup, make up an f, an h^* and a g and plot them on a graph of y vs x (assume $p = 1$). Indicate the sources of error on this plot (see last question). Which source of error is missing from the picture? Why?



Ignorance error is missing because it arises from unobserved variables.

(w) [easy] What is a null model g_0? What data does it make use of? What data does it not make use of?

g_0 is the base case model or the null model that uses no historical data.

(x) [easy] What is a parameter in $\mathcal{H}$?

A parameter specifies a particular value within a function within H.

(y) [easy] Regardless of your answer to what $\mathcal{Y}$ was above in (g), we now coerce $\mathcal{Y} = \{0, 1\}$. What would the null model g_0 be and why?

$g_0 = \text{mode}(Y)$. Measuring which output occurs the most as a prediction without any inputs.

(z) [easy] Regardless of your answer to what $\mathcal{Y}$ was above in (g), we now coerce $\mathcal{Y} = \{0, 1\}$. If we use a threshold model, what would $\mathcal{H}$ be? What would the parameter(s) be?

$H = \{ \mathbf{1}_{(x \geq \theta)} \ : \ \theta \in X \}$.

(aa) [easy] Give an explicit example of g under the threshold model.

g(x) = 1(x > 1).

Problem 4

As alluded to in class, modeling is synonymous with the entire enterprise of science.

In 1964, Richard Feynman, a famous physicist and public intellectual with an inimitably captivating presentation style, gave a series of seven lectures in 1964 at Cornell University on the "character of physical law". Here is a 10min excerpt of one of these lectures about the scientific method. Feel free to watch the entire clip, but for the purposes of this class, we are only interested in the following segments: 0:00-1:00 and 3:48-6:45.

(a) [harder] According to Feynman, how does the scientific method differ from learning from data with regards to building models for reality? (0:08)

Feynman describes the scientific method as starting with a hypothesized law or model, then computing its consequences and comparing them to experiment.

Learning from data begins with data first, using empirical methods to approximate relationships without knowing the true underlying law in advance.

(b) [harder] He uses the phrase "compute consequences". What word did we use in class for "compute consequences"? This word also appears in your diagram in 2a. (0:14)

Prediction/simulation

(c) [harder] When he says compare consequences to "experiment", what word did we use in class for "experiment"? This word also appears in your diagram in 2a. (0:29)

Data.

(d) [harder] When he says "compare consequences to experiment", which part of the diagram in 2a is that comparison?

This corresponds to model validation: comparing predictions to data.

(e) [difficult] When he says "if it disagrees with experiment, it's wrong" (0:44), would a data scientist agree/disagree? What would the data scientist further comment?

They would say that disagreement with data does not automatically make it wrong. Instead it may mean that more data is needed, or that there arent enough proxies.

(f) [difficult] [You can skip his UFO discussion as it belongs in a class on statistical inference on the topic of H_0 vs H_a which is *not* in the curriculum of this class.] He then goes on to say "We can disprove any definite theory. We never prove [a theory] right... We can only be sure we're wrong" (3:48 - 5:08). What does this mean about models in the context of our class?

It means that models cannot be proven true, they are approximations, and can be shown to fail under testing.

(g) [difficult] Further he says, "you cannot prove a *vague* theory wrong" (5:10 - 5:48). What does this mean in the context of mathematical models and metrics?

When a theory is vague, it lacks precise definitions, and metrics, making testing difficult.

(h) [difficult] He then he continues with an example from psychology. Remeber in the 1960's psychoanalysis was very popular. What is his remedy for being able to prove the vague psychology theory right (5:49 - 6:29)?

Feynman's remedy was to make the theory precise by defining clear measurements and testable predictions, turning the vague model into a precise one.

(i) [difficult] He then says "then you can't claim to know anything about it" (6:40). Why can't you know anything about it?

Because without measurable predictions, the theory provides no information about reality.

Just to demonstrate that this modeling enterprise is all over science (not just Physics), I present to you the controversial theoretical political scientist John Mearsheimer. He's all over youtube and there's nothing special about this video that I will link here about Can China Rise Peacefully? Feel free to watch the entire clip, but for the purposes of this class, we are only interested in the following segments referenced in the questions which has nothing to do with China, only his theory of "power politics".

(j) [difficult] Is Mearsheimer's model of great power politics / international relations (i.e., modern history) 9:35-17:22 simple or complicated? Explain.

Mearsheimer's model is simple. It removes many variables to focus on a small number of variables deemed important, building the model off those.

(k) [difficult] Summarize his ideas about limitations of his theory from 39:18-40:00 using vocabulary from this class.

Mearsheimer's model contains model misspecification and ignorance error. His theory omits many covariates and is not intended to make precise predictions, only to explain broad patterns. As such, it is useful for explanation but limited for prediction.